Question 1 *Fill in the proof template for **Direct Proofs** below with the appropriate terms.*

Claim: $p \to q$
Proof. *Assume* $\boxed{?}$
.
.
Thus $\boxed{?}$

Question 2 *Choose the book's recommended sequence for the following steps when writing an **if-then** proof.*

- *Step 1: Unravel the definitions usually found in the assumption and the conclusion of the claim (forward from the beginning of the proof and backward from the end of the proof)*
- *Step 2: Write the last sentence of the proof (restate the conclusion)*
- *Step 3: Assume (restate) the hypothesis*
- *Step 4: Figure out what is missing in linking the two halves of the arguments*

Multiple Choice:

(a) $1 \to 2 \to 3 \to 4$

(b) $3 \to 2 \to 1 \to 4$

(c) $2 \to 1 \to 4 \to 3$

(d) $3 \to 2 \to 4 \to 1$

Question 3 *Which of the following would you need to prove an **if-and-only-if** proof? (Claim:* $p \leftrightarrow q$*)*

Multiple Choice:

(a) $p \to q$

(b) p

(c) p and q

(d) $p \to q$ and $q \to p$

(hint: see proof template 2 for if-and-only-if theorem on page 21)

Question 4 *In this question, we will walk you through an if-then proof. Fill in the missing steps!*

Claim: Suppose a,b,d,x and y are integers. If d|a and d|b, then d|(ax+by).

Proof. Assume d|a and d|b. [given]
From d|a, we can say that $\underbrace{\hspace{2cm}}_{answer1}$
Similarly, from d|b, we can say that $\underbrace{\hspace{2cm}}_{answer2}$

Then ax + by = $\underbrace{\hspace{2cm}}_{answer3}$
= d(mx) + d(ny),
= d(mx + ny), $\underbrace{\hspace{2cm}}_{answer4}$

By definition of divides, $\underbrace{\hspace{2cm}}_{answer5}$

- *Answer 1*

  Multiple Choice:

  (a) ax = d
  (b) there exists an integer m, such that a = dm
  (c) a = dx

- *Answer 2*

  Multiple Choice:

  (a) there exists an integer m, such that b = dm
  (b) there exists an integer n, such that b = dn
  (c) there exists a positive number n, such that bn = d
  (d) b = dy

- *Answer 3*

  Multiple Choice:

  (a) *(dm)x + (dn)y*
  (b) $m + n$
  (c) *dx + dy*
  (d) *am + bn*

- *Answer 4*

  Multiple Choice:

  (a) *m and n are both positive, mx + ny is also positive*
  (b) *since m, n, x, y are integers, mx + ny belongs to* $\mathbb{Z}$
  (c) *Leave it blank [no step necessary]*

- *Answer 5*

  Select All Correct Answers:

  (a) $d|(ax+by)$
  (b) $d|(a+b)$
  (c) $(ax+by)|d$
  (d) *Restate the conclusion part of the if-then claim.*